

A large, detailed photograph of the U.S. Capitol dome, showing its ornate architecture, columns, and the Statue of Freedom on top. The dome is set against a blue sky with scattered white clouds. The image is partially framed by a blue geometric shape on the left and a grey shape at the bottom.

Enterprise Data Inventories

Agencies face challenges and opportunities to increase the value of data assets when implementing data inventories

This report was developed by the Chief Data Officer Council's Data Inventory Working Group to support implementing the Federal Data Strategy and the Foundations for Evidence-based Policymaking Act.

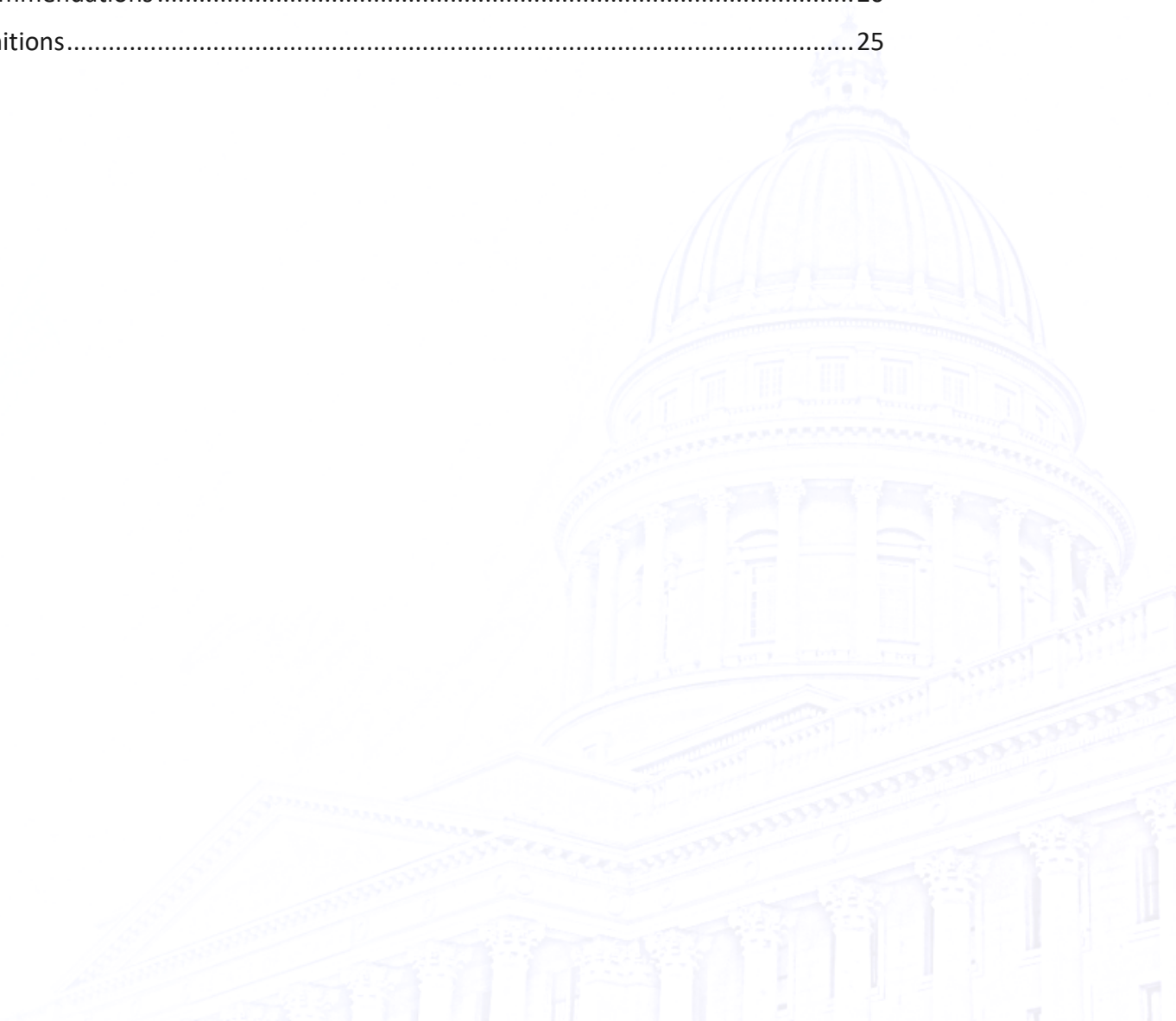
Released April 2022

A decorative graphic element at the bottom of the page consisting of a series of parallel, light blue diagonal lines.



Contents

1. Introduction	1
2. Enterprise Data Inventory Value	4
3. Agency Data Inventory Policy	6
4. Agency Culture	8
5. Technology	12
6. Data Protection and Cyber Security	16
7. Recommendations	20
8. Definitions	25



Acknowledgments

The Federal Chief Data Officers Council Data Inventory developed this report and included input from the CDO Council.

Published: April, 2022

Report Lead: Tod Dabolt, Chief Data Officer, Department of the Interior

Contributors: Anne Levine (FCC), Robin Rappaport (IRS), Arun Acharya (CPSC), Harold Saintelien (Census), Lajuan Bryan-Beveridge (DOC), Chris Alvarez (USDA), Cynthia Parr (ARS), Brian Raub (IRS), Brian Reichenbach (DOS), Hyon Kim (GSA), David Carter (DOI)

Contributing Agencies:

- U.S. Census Bureau
- U.S. Consumer Product Safety Commission
- U.S. Federal Communications Commission
- U.S. Department of Agriculture
- U.S. Department of the Interior
- U.S. Department of State
- U.S. Department of Transportation
- U.S. Department of the Treasury
- U.S. Department of Veterans Affairs
- U.S. General Services Administration
- U.S. Office of Management and Budget
- U.S. Environmental Protection Agency

With thanks to the leadership of the CDO Council Executive Committee

- | | |
|--|--|
| • Ted Kaouk, Chair | • Matthew Graviss, Data Skills and Workforce Development Working Group |
| • Dan Morgan, Vice Chair | • Nikolaos Ipiotis, Data Sharing Working Group |
| • Maria Roat, OMB E-Gov | • Tod Dabolt, Data Inventory Working Group |
| • Dominic Mancini, OMB OIRA | • Melanie Carter, Operations Working Group |
| • Kshendra Paul, Large Agency Committee | |
| • Kirsten Dalboe, Small Agency Committee | |

1. Introduction

The *Foundations for Evidence-Based Policymaking Act of 2018 (Evidence Act)* calls for a systematic rethinking of how the federal government manages and uses the information it collects, emphasizing strong agency coordination for the strategic use of data. Specifically, Title II of the *Evidence Act* is the *Open, Public, Electronic and Necessary (OPEN) Government Data Act* (an Act within an Act), which lays out certain agency responsibilities, including the requirement for a comprehensive data inventory.

The Chief Data Officer's Council Data Inventory Working Group developed this paper to highlight the value proposition for data inventories and describe challenges agencies may face when implementing and managing comprehensive data inventories. It identifies opportunities agencies can take to overcome some of these challenges and includes a set of recommendations directed at Agencies, OMB, and the CDO Council (CDOC).

A data inventory is foundational to any formal data management program. Data inventories enable stakeholders to efficiently find, access and use data assets. Inventories are also indispensable to managers who need to evaluate the extent to which the organization's data helps meet mission goals or can be shared with other agencies to meet their missions. Policy and program managers not only need to know *what* data the organization collects and maintains but also *how* it aligns with its intended purpose, what condition it is in, how it is stored and accessed. In addition, data inventories can inform records management teams who need to understand the record retention schedule of the organization's data and information; security and privacy teams who need to ensure the organization has the appropriate security and access management procedures in place; and legal teams who need to ensure the organization is using the data appropriately.

The purpose of the comprehensive data inventory is to ensure that agency staff and the public have a clear comprehensive understanding of the data assets in the possession of the agency. While OMB data inventory guidance is forthcoming, the statute requires each agency: "shall, to the maximum extent practicable, develop and maintain a comprehensive data inventory that accounts for all data assets created by, collected by, under the control or direction of, or maintained by the agency." Furthermore, the *Evidence Act* requires that the data inventory metadata schema include, to the maximum extent practicable, the following:

- Description of the data asset (including all variable names and definitions).
- Name or title of the data asset.
- Indication of whether the data asset is eligible for disclosure or if it is exempt (or partially exempt) from public disclosure (or the date by which the agency will make such determination).
- Agency or sub-agency responsible for maintaining the data asset.
- Owner of the data asset.
- Date on which the data asset was most recently updated.
- Any use restrictions on the data asset.
- Location of the data asset.

- Method by which the public may access or request access to the data asset.
- Any metadata necessary to make the inventory useful to the agency and the public.

The differences between a Data Inventory and a Data Catalog.

Although the terms “data catalog” and “data inventory” are often used synonymously, they mean very different things. A data catalog is the mechanism that helps users discover the data assets that are found in the data inventory. The data catalog contains such information as the organizational ownership of data assets, its’ meaning to the organization (business metadata), and where and how to access it. A data inventory can contain more technical and granular metadata such as the definitions of specific data elements, their format, valid values, and their completeness. While the concepts are distinct, they are complementary. A user may search a data catalog to find data sets related to a specific topic of interest and then use information in the data inventory to discover the meaning of data elements within the datasets they discovered through the catalog. A data catalog and data inventory should work in tandem to provide organizations with the best value. An item in the data catalog may reference multiple items in the inventory and vice versa.

Starting with basic business metadata, inventories may also link data elements and data assets to additional information regarding security, privacy, and records. Many enterprise data inventories also capture operational metadata. Operational metadata describes the data lineage, data currency, and the flow of the data across an organization. The technical and operational metadata in an inventory change frequently and require automation while the business information associated with the data catalog tends to be more static.

Once inventoried, the data assets described in a comprehensive data inventory must, as per statute, be submitted for inclusion in the Federal Data Catalogue and must be updated in real-time. Also, as per statute, the data assets themselves must be made available to the public via electronic hyperlink or other means of access, unless there are valid reasons for not doing so, as defined under statute. Valid reasons for data assets being exempt (or partially exempt) from public disclosure, are based largely on (but are not limited to), the following criteria:

- risks and restrictions related to the disclosure of personally identifiable information, including when combined with other information
- security considerations, including when combined with other information
- the cost and benefits to the public of converting the data into a usable format,
- risk of legal liability,
- intellectual property rights,
- confidential business information,
- restrictions based on contract or other binding, written agreement,
- consultation with the holder of a right to such data asset.

These reasons for exemption apply to all individual data assets in isolation, but also if, when combined with other available information, the individual data asset may pose such a risk.

***OMB Guidance on the Foundations for Evidence-Based Policymaking Act*¹**

In September 2020, OMB circulated Draft Phase 2 Implementation Guidance for Inter-Agency Review. OMB Draft Guidance recognized that the comprehensive data inventory of each agency will need to be built out over time, reflecting the priorities of the individual agency. Governing and managing the development and operation of a successful data inventory requires a cross-disciplinary team supported by the agency's data governance body.

“Because developing and maintaining a comprehensive data inventory will require managing a significant breadth and depth of data assets, agencies may want to consider leveraging Capital Planning and Investment Control Coordinators, Chief Information Officers, Chief Freedom of Information Act Officers, Controlled Unclassified Information Senior Agency Officials, Enterprise Architects, Senior Agency Officials for Privacy, and Senior Agency Officials for Records Management, as part of the inventory team.”²

Draft OMB Guidance suggests identifying individual assets from a variety of different sources already institutionalized in most agencies, including, but not limited to the following:

- Information Collection Review (ICR), Inventory comprised of ICR requests submitted to OMB under the Paperwork Reduction Act (as listed on [reginfo.gov](https://www.reginfo.gov)).
- Information System Inventory (IT systems).
- Any data assets that are publicly disseminated on an agency website.
- Any data asset that is associated with a system of records as defined by the Privacy Act (SORN).

¹ Phase 2 Guidance

² Draft OMB Phase 2 Guidance